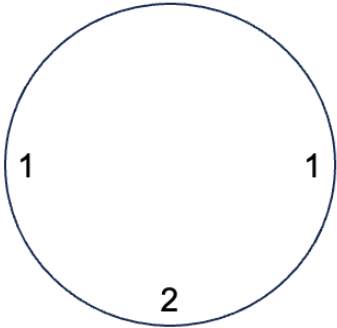


		work is done by the gas	A1
3	(a)	The periodic motion of an object about the equilibrium point in which its <u>acceleration is proportional to the displacement of the object from the equilibrium point</u> and always acts in the <u>opposite direction to its displacement</u> .	B1
	(b)	(i) $\omega = 2\pi f = 2\pi(12) = 75 \text{ rad s}^{-1}$	C1
		(ii) Amplitude $x_0 = 32 / 2 = 16 \text{ mm} = 0.016 \text{ m}$ $v_0 = \omega x_0 = (75.4)(0.016) = 1.2 \text{ m s}^{-1}$ Obtain zero mark if the first M1 mark is not obtained, ie using the incorrect amplitude.	M1 A1
		(iii) Maximum contact force at lowest point $F_{\text{net}} = N - W = m\omega^2 x_0$ $N = m\omega^2 x_0 + mg$ $= (0.025)(75.4^2)(0.016) + (0.025)(9.81)$ $= 2.5 \text{ N}$	M1 A1

3

		(iv)	Front view of disc  1 mark each Note: just need to mark and label one of the "1"	M2
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4	(c)	(i)	18.2 cm	A1
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